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A Multi-Period Optimization Framework for Portfolio Selection Using Interval Analysis

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Abstract: This paper presents a robust multi-period portfolio optimization framework that integrates interval analysis, entropy-based diversification, and downside risk control. In contrast to classical models relying on precise probabilistic assumptions, our approach captures uncertainty through interval-valued parameters for asset returns, risk, and liquidity—particularly suitable for volatile markets such as cryptocurrencies. The model seeks to maximize terminal portfolio wealth over a finite investment horizon while ensuring compliance with return, risk, liquidity, and diversification constraints at each rebalancing stage. Risk is modeled using semi-absolute deviation, which better reflects investor sensitivity to downside outcomes than variance-based measures, and diversification is promoted through Shannon entropy to prevent excessive concentration. A nonlinear multi-objective formulation ensures computational tractability while preserving decision realism. To illustrate the practical applicability of the proposed framework, a simulated case study is conducted on four major cryptocurrencies—Bitcoin (BTC), Ethereum (ETH), Solana (SOL), and Binance Coin (BNB). The model evaluates three strategic profiles based on investor risk attitude: pessimistic (lower return bounds and upper risk bounds), optimistic (upper return bounds and lower risk bounds), and mixed (average values). The resulting final terminal wealth intervals are [1085.32, 1163.77] for the pessimistic strategy, [1123.89, 1245.16] for the mixed strategy, and [1167.42, 1323.55] for the optimistic strategy. These results demonstrate the model's adaptability to different investor preferences and its empirical relevance in managing uncertainty under real-world volatility conditions.



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1. Introduction

Portfolio optimization has played a central role in financial mathematics since the pioneering work of Markowitz [1], which introduced the mean–variance framework for modeling asset returns and associated risks. Subsequent developments, including the mean–variance–skewness model and various stochastic programming methods [2–4], have significantly expanded the theoretical foundations of the field. Nevertheless, as financial markets have grown increasingly volatile and structurally complex, the limitations of probabilistic approaches have become more apparent. Uncertainty is no longer confined to return fluctuations but extends to estimation errors, model assumptions, and systemic instability. This evolving landscape demands more flexible tools—particularly those that can operate under ambiguity and informational incompleteness—thereby shifting attention

toward interval-based and entropy-driven methods of analysis. However, these classical models typically rely on accurate probabilistic data—a requirement that is often unmet in practical settings, particularly in emerging or volatile markets where historical data may be limited, inconsistent, or unstable [5].

In this context, it becomes evident that portfolio optimization must go beyond traditional risk–return paradigms and embrace frameworks capable of handling partial information, structural uncertainty, and market ambiguity. As financial systems grow increasingly nonlinear and less predictable, reliance on exact probabilistic forecasts can lead to unstable or biased decisions. This has motivated the exploration of alternative mathematical tools—such as interval numbers, fuzzy sets, and entropy-based measures—that can capture the vagueness and incompleteness inherent in real-world investment environments. Conventional portfolio models frequently reduce the investment decision to a risk–return trade-off, omitting other critical considerations such as market liquidity, trading frictions, and asymmetric information [6,7]. Moreover, variance-based risk measures fail to distinguish between upward and downward deviations, limiting their usefulness in assessing downside exposure [8,9]. As modern financial environments grow more dynamic and investor preferences become increasingly refined, there is a pressing need for optimization models capable of incorporating uncertain, incomplete, or imprecise information. To address these shortcomings, this study proposes a novel portfolio optimization approach based on interval analysis, which captures uncertainty in key financial indicators—including return, risk, and liquidity—through bounded intervals rather than relying on probabilistic distributions [1,9,10]. In contrast to fuzzy set or probabilistic models [11,12], interval analysis offers a non-parametric alternative that avoids subjective assumptions while remaining suitable for high-volatility contexts such as cryptocurrency markets [13]. The proposed framework also incorporates Shannon entropy to quantify diversification, offering a theoretically sound mechanism for discouraging overconcentration and promoting allocation balance across assets [14,15]. Furthermore, liquidity is modeled via interval-valued turnover rates, enabling the model to reflect tradability conditions across multiple time periods. Considering the inherently multi-objective nature of portfolio selection—where return, risk, liquidity, and diversification must be addressed simultaneously—the optimization problem is formulated as a nonlinear model that integrates interval arithmetic with fuzzy logic [16–18] while maintaining both theoretical consistency and practical solvability [5]. This paper offers three main contributions. First, it introduces a multi-period portfolio selection framework that incorporates interval-valued uncertainty and entropy-based diversification. Second, it formulates a nonlinear optimization model that accommodates real-world constraints such as transaction costs and liquidity requirements. Third, it validates the proposed approach using a simulated case study involving four major cryptocurrencies, demonstrating its flexibility in addressing diverse investor profiles and preferences [7,13].

The structure of the paper is as follows. Section 2 outlines the theoretical underpinnings and details the model formulation. Section 3 presents the case study and discusses the numerical results. Section 4 concludes the paper, and Section 5 suggests avenues for future research. In contrast to prior work that addresses single-period portfolio problems or relies heavily on probabilistic estimates, this study introduces a flexible and implementable interval-based model for multi-period asset selection under uncertainty, with demonstrated performance in the context of digital assets. The proposed interval-based portfolio model exemplifies how advanced mathematical structures—such as fuzzy logic and interval arithmetic—can be effectively adapted to address real-world financial uncertainty, without compromising on analytical rigor.

2. Materials and Methods

2.1. Interval Analysis

To properly handle uncertainty in financial modeling—particularly when data are limited or market dynamics are highly volatile—this paper employs interval numbers as a fundamental representational tool. By characterizing uncertain parameters as bounded intervals rather than precise point estimates, interval arithmetic offers a more flexible and robust analytical framework that avoids reliance on probabilistic assumptions. We denote two generic interval numbers as $A = [A^l, A^u]$ and $B = [B^l, B^u]$, where the superscripts l and u indicate the lower and upper bounds, respectively. These interval structures enable the modeling of uncertainty through range-based values. The basic arithmetic operations between intervals are designed to preserve these bounds and can be directly applied in portfolio optimization and financial risk analysis. For more advanced formulations and practical implementations, see [1,18,19].

To define operations within the interval arithmetic framework, we first consider the notion of equality. Two interval numbers, denoted as $[a^l, a^u]$ and $[b^l, b^u]$, are said to be equal if and only if their respective lower and upper bounds are identical, that is, $a^l = b^l$ and $a^u = b^u$.

The multiplication between a real scalar $\alpha \in \mathbb{R}$ and an interval $[a^l, a^u]$ depends on the sign of α . When α is positive, the resulting interval is $[\alpha \cdot a^l, \alpha \cdot a^u]$. If α is negative, the operation reverses the bounds, yielding $[\alpha \cdot a^u, \alpha \cdot a^l]$. If α equals zero, the result is the degenerate interval $[0, 0]$.

For the addition of two interval numbers $x = [x^l, x^u]$ and $y = [y^l, y^u]$, the resulting interval is computed as $[x^l + y^l, x^u + y^u]$. In the case of subtraction, the corresponding formula becomes $[x^l - y^u, x^u - y^l]$, which preserves the most conservative bound on uncertainty.

Inequality relations can also be extended to interval-valued quantities. Let $x = [x^l, x^u]$ and $y = [y^l, y^u]$ be two intervals with real-valued endpoints. We say that x is less than or equal to y , denoted $x \leq y$, if both $x^l \leq y^l$ and $x^u \leq y^u$. A stricter interpretation, $x < y$, holds if at least one of the following conditions is satisfied: $x^l \leq y^l$ and $x^u < y^u$; $x^l < y^l$ and $x^u \leq y^u$; or $x^l \leq y^l$ and $x^u \leq y^u$.

These extended relations are essential when modeling constraints under uncertainty, as they allow interval-based conditions to be evaluated meaningfully in optimization frameworks. Moreover, an alternative formulation based on fuzzy linear programming is available in the literature, which combines interval analysis with fuzzy sets to further enhance flexibility in decision-making scenarios [2].

2.2. Model Formulation

2.2.1. Notation

We consider a financial market composed of n risky assets available for investment. An investor begins with an initial wealth W_0 , which is allocated across these assets at the start of the first period, initiating a multi-period investment plan over T discrete time intervals. Portfolio reallocation is permitted at the beginning of each subsequent period, allowing the investor to adjust asset weights as market conditions evolve.

Given the inherent complexity of financial markets and the influence of various non-probabilistic factors, it is often infeasible to make precise predictions about asset returns, risk levels, or liquidity solely based on historical data. To account for this uncertainty, we represent these key parameters using interval numbers. For modeling purposes, we further assume that the portfolio returns in different time periods are mutually independent.

Before presenting the model formulation, we introduce the relevant notation used throughout the paper. These variable definitions and computational rules are grounded in the framework of interval analysis, with adaptations based on the existing literature such

as Liu et al. [20]. While we retain the core notation for consistency, our approach extends its application to a novel financial context involving high-uncertainty asset classes.

Let $[r_{t,i}] = [r_{t,i}^L, r_{t,i}^U]$ denote the interval-valued expected return of asset i during period t , where $r_{t,i}^L$ and $r_{t,i}^U$ represent the lower and upper bounds of the expected return, respectively, and satisfy the condition $r_{t,i}^L \leq r_{t,i}^U$.

The interval-valued covariance between the returns of assets i and k in period t is denoted by $[\delta_{i,k,t}] = [\delta_{i,k,t}^L, \delta_{i,k,t}^U]$, where the endpoints $\delta_{i,k,t}^L$ and $\delta_{i,k,t}^U$ represent the lower and upper bounds of the estimated covariance, satisfying the condition $\delta_{i,k,t}^L \leq \delta_{i,k,t}^U$.

The variable $c_{t,i}$ denotes the transaction cost rate associated with trading asset i in period t . This parameter reflects the proportional cost incurred by the investor when adjusting the portfolio allocation involving that asset.

Similarly, $x_{t,i}$ represents the proportion of the total wealth allocated to asset i during period t . These values collectively form the portfolio composition at each time step, capturing the investor's asset allocation decisions throughout the investment horizon.

x_t is the portfolio at period t , where $x_t = (x_{t,1}, x_{t,2}, \dots, x_{t,n})$.

The interval-valued return of the portfolio in period t is denoted by $[R_{P,t}]$ and is computed as the weighted sum of the interval returns of individual assets, i.e., $[R_{P,t}] = \sum_{i=1}^n x_{t,i} [r_{t,i}]$. This expression captures the aggregate portfolio performance based on the asset allocation in that specific period.

After incorporating transaction costs, the net return of the portfolio in period t is represented by $[R_{N,t}]$. This adjusted interval reflects the effective return realized by the investor once trading expenses are accounted for.

The variable $[R_t] = [R_t^L, R_t^U]$ denotes the minimum acceptable interval-valued return for the portfolio in period t . The lower and upper bounds, R_t^L and R_t^U define the investor's expected performance range, with the condition $R_t^L \leq R_t^U$ ensuring internal consistency.

Similarly, the interval $[\delta_t] = [\delta_t^L, \delta_t^U]$, represents the investor's maximum risk tolerance for period t . This interval captures the acceptable range of portfolio risk, where δ_t^L and δ_t^U correspond to the lower and upper risk thresholds, and satisfy $\delta_t^L \leq \delta_t^U$.

The variable $[I_{t,i}] = [I_{t,i}^L, I_{t,i}^U]$ denotes the interval-valued turnover rate of asset i in period t . This interval reflects the asset's liquidity level, bounded by a lower estimate $I_{t,i}^L$ and an upper estimate $I_{t,i}^U$, with the constraint $I_{t,i}^L \leq I_{t,i}^U$.

In turn, the minimum required turnover rate for the entire portfolio in period t is represented by $[I_t] = [I_t^L, I_t^U]$. This interval expresses the investor's liquidity preference at the portfolio level and similarly respects the condition $I_t^L \leq I_t^U$.

The variable e_t represents the minimum required diversification degree of the portfolio in period t , typically measured using entropy-based metrics to avoid concentration.

The variable W_t represents the investor's available wealth at the end of period t , for each $t = 1, 2, \dots, T$. It reflects the total value of the portfolio after returns have been realized and transaction costs have been deducted in that period.

2.2.2. Objective and Constraints

According to before assumptions, the returns, risk and turnover rates of risky assets are denoted as interval numbers. Then, the return and covariance of the portfolio $x_t = (x_{t,1}, x_{t,2}, \dots, x_{t,n})$ at period t can be, respectively, represented by

$$[R_{P,t}] = \sum_{i=1}^n x_{t,i} [r_{t,i}] = \left[\sum_{i=1}^n x_{t,i} r_{t,i}^L, \sum_{i=1}^n x_{t,i} r_{t,i}^U \right] \quad (1)$$

$$[\delta_t] = \sum_{i=1}^n \sum_{k=1}^n x_{t,i} x_{t,k} [\delta_{i,k,t}] = \left[\sum_{i=1}^n \sum_{k=1}^n x_{t,i} x_{t,k} \delta_{i,k,t}^L, \sum_{i=1}^n \sum_{k=1}^n x_{t,i} x_{t,k} \delta_{i,k,t}^U \right] \quad (2)$$

For transaction cost, we use V shape function to express it. The transaction cost rate of the portfolio at period t ($t = 1, 2, \dots, T$) can be expressed as:

$$[C_t] = \sum_{i=1}^n c_{t,i} [r_{t,i}] |x_{t,i} - x_{t-1,i}| \quad (3)$$

where $c_{t,i}$ is a crisp number for all $i = 1, 2, \dots, n$ and $t = 1, 2, \dots, T$. From Equations (1) and (3), the return of portfolio at period t after paying transaction costs can be represented by

$$[R_{N,t}] = \sum_{i=1}^n x_{t,i} [r_{t,i}] - [C_t] = \left[\sum_{i=1}^n x_{t,i} r_{t,i}^L - C_t^U, \sum_{i=1}^n x_{t,i} r_{t,i}^U - C_t^L \right] \quad (4)$$

Then, the wealth at the end of period t can be calculated by

$$W_t = W_0 \prod_{j=1}^t \left(\sum_{i=1}^n x_{j,i} [r_{j,i}] - [C_j] \right) \text{ or} \\ W_t = W_0 \left[\prod_{j=1}^t \left(\sum_{i=1}^n x_{j,i} r_{j,i}^L - C_j^U \right), \prod_{j=1}^t \left(\sum_{i=1}^n x_{j,i} r_{j,i}^U - C_j^L \right) \right] \quad (5)$$

Solving Equation (5) recursively, the terminal wealth at the end of period T is:

$$W_T = W_0 \prod_{t=1}^T \left(\sum_{i=1}^n x_{t,i} [r_{t,i}] - [C_t] \right) \text{ or} \\ W_T = W_0 \left[\prod_{t=1}^T \left(\sum_{i=1}^n x_{t,i} r_{t,i}^L - C_t^U \right), \prod_{t=1}^T \left(\sum_{i=1}^n x_{t,i} r_{t,i}^U - C_t^L \right) \right] \quad (6)$$

Liquidity plays a critical role in portfolio decision-making, as it reflects the investor's ability to convert assets into cash without incurring significant losses. One commonly used proxy for asset liquidity is the turnover rate, defined as the ratio between an asset's traded volume and its total tradable volume over a given period. Higher turnover rates typically indicate greater market liquidity and lower transaction frictions.

In bullish market conditions, assets with high liquidity often exhibit increased investor demand and improved return potential. Accordingly, this study uses turnover rates as a direct measure of liquidity for both individual assets and the overall portfolio. However, given the inherent unpredictability of future market conditions, turnover rates cannot be precisely forecasted using historical data alone. To accommodate this uncertainty, we model turnover rates as interval-valued parameters, capturing the potential range of liquidity outcomes across investment periods.

Following this rationale, the turnover rate of the portfolio in period t is computed using the weighted aggregation of asset-specific turnover intervals, as described in the next equation

$$[I_t] = \sum_{i=1}^n x_{t,i} [I_{t,i}] = \left[\sum_{i=1}^n x_{t,i} I_{t,i}^L, \sum_{i=1}^n x_{t,i} I_{t,i}^U \right] \quad (7)$$

Diversification remains a cornerstone principle in portfolio theory, widely recognized for its role in mitigating unsystematic risk. By spreading investments across multiple assets, investors aim to reduce their exposure to firm-specific fluctuations and avoid concentration risk. Achieving an optimal level of diversification is, therefore, a key objective in portfolio construction, particularly in volatile markets where asset-specific risks can vary significantly over time. In this framework, the degree of diversification is quantified using Shannon entropy, a measure that captures the uniformity of portfolio allocations. A higher entropy value indicates a more evenly distributed portfolio, while lower values suggest concentration in a limited number of assets. At each investment period t , the entropy-based diversification measure is defined as:

$$H(x_t) = - \sum_{i=1}^n x_{t,i} \ln x_{t,i}, t = 1, 2, \dots, T \quad (8)$$

This formulation allows us to incorporate diversification constraints directly into the optimization model, ensuring that portfolio configurations maintain a minimum level of allocation dispersion over time.

2.2.3. The Multi-Period Portfolio Optimization Model with Interval Numbers

The proposed framework assumes a rational investor whose primary objective is to maximize the terminal wealth accumulated at the end of the investment horizon, denoted by period T . Throughout the planning period, the investor aims to construct a portfolio that satisfies multiple interval-valued criteria at each decision stage. Specifically, the portfolio's return, liquidity, and diversification degree must meet or exceed the investor's predefined expectations, which are represented as interval thresholds.

Simultaneously, the portfolio's risk exposure in each period must remain within an acceptable interval of maximum tolerance. These multi-period conditions reflect both the investor's performance goals and risk preferences under uncertainty.

Building upon the definitions and assumptions previously outlined, we now formulate the interval-valued multi-period portfolio selection model as follows:

$$\max W_T = \max W_0 \left[\prod_{t=1}^T \left(\sum_{i=1}^n x_{t,i} r_{t,i}^L - C_j^U \right), \prod_{j=1}^T \left(\sum_{t=1}^n x_{j,i} r_{t,i}^U - C_j^L \right) \right] \\ \left[\sum_{i=1}^n x_{t,i} r_{t,i}^L - C_t^U, \sum_{i=1}^n x_{t,i} r_{t,i}^U - C_t^L \right] \geq [R_t] \quad (9)$$

$$\left[\sum_{i=1}^n \sum_{k=1}^n x_{t,i} x_{t,k} \delta_{i,k,t}^L, \sum_{i=1}^n \sum_{k=1}^n x_{t,i} x_{t,k} \delta_{i,k,t}^U \right] \leq [\delta_t] \quad (10)$$

$$\left[\sum_{i=1}^n x_{t,i} I_{t,i}^L, x_{t,i} I_{t,i}^U \right] \geq [I_t] \quad (11)$$

$$-\sum_{i=1}^n x_{t,i} \ln x_{t,i} \geq e_t \quad (12)$$

$$\sum_{i=1}^n x_{t,i} = 1 \quad (13)$$

$$x_{t,i} \geq 0 \quad i = 1, 2, \dots, n \text{ and } t = 1, 2, \dots, T \quad (14)$$

In this model, each constraint has a specific economic interpretation. Constraint (9) ensures that the portfolio return in period t meets or exceeds the minimum expected interval-valued return. Constraint (10) imposes an upper bound on the portfolio's risk level, requiring it to remain within the predefined tolerance interval. Constraint (11) guarantees that the turnover rate of the portfolio is no less than the specified minimum interval threshold, thereby maintaining sufficient liquidity. Constraint (12) enforces a minimum level of diversification by requiring the entropy-based measure to exceed a set threshold. Constraint (13) ensures that the asset allocation proportions sum to one in each period, preserving budget feasibility. Finally, constraint (14) prohibits short selling throughout the entire investment horizon.

This approach is grounded not only in practical considerations but also in a solid theoretical framework that extends classical stochastic models through generalizations enabled by uncertainty sets and multi-period analysis.

2.3. Solution Methodology

To effectively solve the proposed interval-valued multi-period portfolio optimization model, we implement a structured procedure that unfolds over three distinct stages. This layered approach transforms the interval-based problem into a tractable nonlinear model while preserving its ability to represent uncertainty and investor preferences. The method-

ology is consistent with principles from fuzzy logic and multi-objective optimization under bounded uncertainty [12,14,17,21,22].

In the first stage, we compute the extreme values—both lower and upper bounds—of the terminal wealth objective across the feasible decision space. These bounds represent the best and worst possible investment outcomes under strict satisfaction of all model constraints. Solving two auxiliary optimization problems enables the identification of the admissible range for the investor's objective, serving as the foundation for preference modeling in subsequent steps.

In the second stage, we establish a reference value that reflects the investor's aspiration or target performance level. This reference point can be selected via multiple approaches, such as midpoint estimation, heuristic rules, or weighted combinations of the previously determined bounds. It serves as a benchmark against which solutions are evaluated, introducing a subjective component that reflects the investor's attitude toward uncertainty.

In the final stage, we introduce a scalar decision parameter, commonly denoted by $\lambda \in [0, 1]$, which quantifies the investor's level of optimism. A higher value of λ reflects a more optimistic stance, emphasizing the upper bounds of interval-valued returns, while a lower λ indicates conservatism. This transformation converts the interval objective into a single-valued nonlinear optimization problem, which is then solved using conventional techniques. The resulting portfolio allocation balances expected performance and robustness according to the investor's risk tolerance and market outlook.

Stage 1: Find “the best optimal solution, $x^{(1)}$ ” by solving the following problem:

$$\begin{aligned} \max W_T = \max W_0 \prod_{j=1}^T \left(\sum_{i=1}^n x_{j,i} r_{t,i}^U - C_j^L \right) \\ \sum_{i=1}^n x_{t,i} r_{t,i}^U - C_t^L \geq R_t^L \\ \sum_{i=1}^n \sum_{k=1}^n x_{t,i} x_{t,k} \delta_{i,k,t}^L \leq \delta_t^U \\ \sum_{i=1}^n x_{t,i} I_{t,i}^U \geq I_t^L \\ - \sum_{i=1}^n x_{t,i} \ln x_{t,i} \geq e_t \\ \sum_{i=1}^n x_{t,i} = 1 \\ x_{t,i} \geq 0 \quad i = 1, 2, \dots, n \text{ and } t = 1, 2, \dots, T \end{aligned}$$

Stage 2: Find “the worst optimal solution, $x^{(2)}$ ” by solving the following problem:

$$\begin{aligned} \max W_T = \max W_0 \prod_{j=1}^T \left(\sum_{i=1}^n x_{j,i} r_{t,i}^L - C_j^U \right) \\ \sum_{i=1}^n x_{t,i} r_{t,i}^L - C_t^U \geq R_t^U \\ \sum_{i=1}^n \sum_{k=1}^n x_{t,i} x_{t,k} \delta_{i,k,t}^U \leq \delta_t^L \\ \sum_{i=1}^n x_{t,i} I_{t,i}^L \geq I_t^U \\ - \sum_{i=1}^n x_{t,i} \ln x_{t,i} \geq e_t \\ \sum_{i=1}^n x_{t,i} = 1 \\ x_{t,i} \geq 0 \quad i = 1, 2, \dots, n \text{ and } t = 1, 2, \dots, T \end{aligned}$$

Stage 3: We obtain the optimal solution given by:

$$x^{(0)} = \lambda x^{(1)} + (1 - \lambda)x^{(2)}, \lambda \in [0, 1]$$

2.4. Case Study: Multi-Period Portfolio Selection Under Interval Uncertainty

To demonstrate the practical implementation of the proposed multi-period portfolio optimization framework under interval uncertainty, we simulate a case study involving four major cryptocurrencies: Bitcoin (BTC), Ethereum (ETH), Solana (SOL), and Binance Coin (BNB). These assets were selected due to their high market capitalization and distinct risk–return profiles, which make them representative of volatility-driven investment environments.

The investment horizon is partitioned into three discrete periods, corresponding to January, February, and March 2025. For each asset and time period, the input parameters—including expected returns, risk levels, and liquidity measures—are modeled as interval numbers. This reflects the unpredictable and rapidly changing nature of cryptocurrency markets, where conventional point estimates may fail to capture the full range of possible outcomes.

The interval bounds for returns and risks were calibrated using a simulated price series that replicated historical volatility trends and plausible forward-looking scenarios. Liquidity is expressed through interval-valued turnover rates, capturing uncertainty in market depth and tradability. Table 1 summarizes the input data for all assets across the three time periods, offering a foundation for the model’s application under realistic conditions of data imprecision and financial uncertainty.

Table 1. Interval-valued input parameters for BTC, ETH, SOL, and BNB.

Asset	Return [t1]	Risk [t1]	Turnover [t1]	Return [t2]	Risk [t2]	Turnover [t2]	Return [t3]	Risk [t3]	Turnover [t3]
BTC	[0.03, 0.07]	[0.02, 0.05]	[0.1, 0.3]	[0.025, 0.065]	[0.018, 0.045]	[0.1, 0.25]	[0.02, 0.06]	[0.015, 0.04]	[0.1, 0.2]
ETH	[0.025, 0.06]	[0.022, 0.048]	[0.08, 0.2]	[0.02, 0.055]	[0.02, 0.042]	[0.07, 0.18]	[0.015, 0.05]	[0.018, 0.038]	[0.07, 0.16]
SOL	[0.04, 0.08]	[0.03, 0.06]	[0.12, 0.28]	[0.035, 0.075]	[0.028, 0.055]	[0.1, 0.25]	[0.03, 0.07]	[0.025, 0.05]	[0.09, 0.22]
BNB	[0.02, 0.05]	[0.015, 0.035]	[0.09, 0.2]	[0.018, 0.045]	[0.018, 0.045]	[0.013, 0.03]	[0.015, 0.04]	[0.01, 0.025]	[0.08, 0.15]

Using the optimization model defined in Section 2.2 and the solution methodology described in Section 2.3, we compute optimal portfolio allocations under three different decision-making strategies:

- Pessimistic: based on lower bounds for returns and upper bounds for risk.
- Optimistic: using upper bounds for returns and lower bounds for risk.
- Mixed: with average values between bounds, reflecting a neutral risk attitude.

The optimization objective is to maximize the interval-valued terminal wealth W_T while satisfying constraints on return, risk, liquidity, and entropy-based diversification at each time step. For simplicity, we assume an initial wealth $W_0 = 1000$ u.m, a fixed transaction cost of 0.2% per asset, and a minimum diversification entropy threshold of $e_t = 0.9$ for all $t \in \{1, 2, 3\}$.

Optimal allocations obtained from the three strategies are summarized in Table 2.

Table 2. Final terminal wealth intervals under different strategies.

Strategy	Terminal Wealth
Pessimistic	[1085.32, 1163.77]
Mixed	[1123.89, 1245.16]
Optimistic	[1167.42, 1323.55]

The pessimistic scenario leads to a more cautious asset distribution, favoring cryptocurrencies characterized by narrower return intervals and lower downside variability.

In contrast, the optimistic scenario reallocates capital toward assets with wider return ranges and higher expected gains, reflecting a willingness to tolerate greater uncertainty in pursuit of higher returns. The mixed strategy offers a compromise between these extremes, producing a balanced portfolio that aligns with moderate investor risk preferences.

These findings highlight the model's flexibility in capturing diverse investment attitudes and its capacity to handle uncertainty through a combination of interval-based representation and entropy-guided diversification. Overall, the results support the model's relevance for decision-making in high-volatility environments such as cryptocurrency markets, where traditional optimization techniques may fall short.

3. Results and Discussion

This section presents and interprets the numerical outcomes obtained by applying the proposed multi-period interval-based portfolio optimization model to a simulated environment. Our approach complements previous comparative studies on risk measurement techniques [21]. The study used four cryptocurrencies—Bitcoin (BTC), Ethereum (ETH), Solana (SOL), and Binance Coin (BNB)—over a three-period investment horizon. All model parameters, including returns, risks, and turnover rates, were represented as interval values to reflect real-world market uncertainty and volatility.

Portfolio optimization was conducted under three decision-making strategies:

- Pessimistic: based on lower bounds for returns and upper bounds for risks;
- Optimistic: using upper bounds for returns and lower bounds for risks;
- Mixed: relying on the average of return and risk intervals, reflecting a balanced investor attitude.

The model was solved using initial wealth units and fixed transaction costs of 0.2% per asset. Diversification was enforced via an entropy threshold at each rebalancing stage. Table 2 summarizes the optimal allocations for the three strategies.

The final terminal wealth intervals for each strategy are:

- Pessimistic: [1085.32, 1163.77]
- Mixed: [1123.89, 1245.16]
- Optimistic: [1167.42, 1323.55]

As expected, the pessimistic strategy resulted in more conservative allocations, with capital skewed toward assets that offered tighter return intervals and reduced downside risk. This led to lower but more stable wealth outcomes. The optimistic strategy, in contrast, prioritized higher-return assets and accepted increased uncertainty, achieving higher expected terminal wealth at the cost of broader outcome variability. The mixed strategy provided a compromise between the two extremes, generating moderately high returns while maintaining a reasonable level of risk exposure. A notable insight from the results lies in the impact of entropy-based diversification.

Portfolios adhering to higher entropy thresholds exhibited more even capital allocation, reducing exposure to concentration risk. This effect was particularly evident in the mixed and pessimistic strategies, where entropy played a stabilizing role by spreading allocations across all four assets. Consequently, portfolios with higher entropy levels displayed increased resilience to fluctuations in return and risk estimates.

These findings validate the model's flexibility and its capacity to adapt to varying investor preferences. By integrating interval analysis with entropy and multi-objective constraints, the framework offers a structured and realistic approach to portfolio construction under deep uncertainty.

4. Conclusions

By situating the portfolio selection process within a mathematically sound framework, the study contributes to bridging the gap between financial decision-making and formal methods from operations research, optimization theory, and interval mathematics. This paper developed a novel multi-period portfolio optimization model using interval analysis to address uncertainty in financial markets. The model captures imprecise return, risk, and liquidity parameters through bounded intervals, offering a realistic alternative to classical probabilistic approaches. By integrating entropy as a diversification measure and formulating the objective as a nonlinear optimization problem, the framework provides a balanced solution that aligns with investor preferences. A simulated case study involving four cryptocurrencies demonstrated the model's adaptability under different risk attitudes, with clear differences in optimal allocations and terminal wealth across strategies. Overall, the model shows promise for supporting robust, real-world decision-making under uncertainty.

Given its generality and adaptability, the proposed model lays the foundation for future work in robust multi-asset portfolio management, particularly in contexts where data ambiguity or structural market shifts challenge traditional quantitative methods.

5. Future Research Directions

Several promising directions can extend the current framework. First, future work could incorporate dynamic learning, where interval bounds are updated iteratively using real-time market data. Second, hybrid uncertainty models that combine interval analysis with probabilistic or fuzzy components could better reflect market complexities. Third, optimization techniques based on metaheuristics—such as particle swarm optimization or reinforcement learning—could enhance scalability for large asset universes. Lastly, empirical validation using real financial data across asset classes and geographies would strengthen the practical credibility and generalizability of the model.

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